

Solutions Term test 3 - Math 1210 - June 5, 2023

1. Consider the matrices

$$A = \begin{pmatrix} 2 & -1 \\ 0 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & -5 \end{pmatrix} \quad C = \begin{pmatrix} 1 & -2 \\ 2 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- (a) (5 points) Evaluate the expression $B^T(A - 2C)$ if it is defined, or explain why it is undefined.

Solution: The expression is well-defined. First we compute

$$A - 2C = \begin{pmatrix} 2 & -1 \\ 0 & -3 \end{pmatrix} - 2 \begin{pmatrix} 1 & -2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ -4 & -3 \end{pmatrix}.$$

Hence

$$B^T(A - 2C) = \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ -1 & -5 \end{pmatrix} \begin{pmatrix} 0 & 3 \\ -4 & -3 \end{pmatrix} = \begin{pmatrix} -8 & -3 \\ -12 & -3 \\ 20 & 12 \end{pmatrix}$$

- (b) (2 points) Evaluate the expression $A^T C - B^2$ if it is defined, or explain why it is undefined.

Solution: The matrix A^T is a 2×2 matrix and C is a 2×2 matrix. Hence $A^T C$ is a well-defined 2×2 matrix. However, the matrix B is a 2×3 matrix. Hence $B^2 = B \cdot B$ is not well-defined since a 2×3 matrix cannot be multiplied with another 2×3 matrix.

- (c) (3 points) Evaluate the expression $2AC - D^T B$ if it is defined, or explain why it is undefined.

Solution: The matrix A is a 2×2 matrix and C is 2×2 so $2AC$ is a well-defined 2×2 matrix. The matrix $D^T = (1, -1)$ is a 1×2 matrix and B is 2×3 so $-D^T B$ is a well-defined 1×3 matrix. But we cannot add a 2×2 matrix and a 1×3 matrix, so the expression is undefined.

- (d) (2 points) Suppose that X is a matrix such that the expression $(BX)^T$ is a well-defined 5×2 matrix. What is the size of X ?

Solution: If $(BX)^T$ is a 5×2 matrix, then BX is a 2×5 matrix. Suppose that $m \times n$ is the size of X . Since B is 2×3 , we need to have $m = 3$ in order to have a well-defined product BX . So we know that X is a $3 \times n$ matrix, and the product BX is $2 \times n$. Since we want this to be 2×5 , we must have $n = 5$. In conclusion, the matrix X is of size 3×5 .

2. Consider the vector $\mathbf{u} = \langle 2m, -3, 1 \rangle$ and $\mathbf{v} = \langle -m, m, 3 \rangle$ where m is a real number.

- (a) (3 points) For which value(s) m are the two vectors $\mathbf{u}$ and $\mathbf{v}$ perpendicular?

Solution: The two vectors are orthogonal precisely when dot product of the two vectors

$$\mathbf{u} \cdot \mathbf{v} = -2m^2 - 3m + 3$$

is equal to 0. Hence we have to solve the quadratic equation $-2m^2 - 3m + 3 = 0$ for m . The discriminant of the equation is $(-3)^2 - 4 \cdot (-2) \cdot 3 = 33$. Hence, the two values of m such that the vectors are orthogonal are $-\frac{3+\sqrt{33}}{4}$ and $-\frac{3-\sqrt{33}}{4}$.

- (b) (3 points) For $m = -1$, find a vector of length 3 in the opposite direction to $\mathbf{u}$.

Solution: When $m = -1$, the vector is $\mathbf{u} = \langle -2, -3, 1 \rangle$. Its length is

$$|\mathbf{u}| = \sqrt{(-2)^2 + (-3)^2 + 1^2} = \sqrt{14}.$$

The vector

$$-3 \frac{\mathbf{u}}{|\mathbf{u}|} = \left\langle \frac{6}{\sqrt{14}}, \frac{9}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle$$

has length 3 and is in the opposite direction to $\mathbf{u}$.

3. (a) (5 points) Find parametric and symmetric equations for the line through the points $P(2, -1, 3)$ and $Q(3, 3, 1)$. Use the symmetric equation to find two planes that intersect along this line.

Solution: We begin with the parametric equation. The vector $\mathbf{PQ} = \langle 1, 4, -2 \rangle$ is parallel to the line. Hence the line is going through $P(2, -1, 3)$ and parallel to $\mathbf{PQ}$ so that parametric equations of L are

$$x = 2 + t, \quad y = -1 + 4t, \quad z = 3 - 2t$$

where t is a real number.

A symmetric equation is given by equating the parametric equation for t :

$$(x - 2) = \frac{y + 1}{4} = -\frac{z - 3}{2}.$$

From the symmetric equations we deduce that the two planes

$$4(x - 2) - (y + 1) = 0$$

and

$$2(y + 1) + 4(z - 3) = 0$$

intersect along the line L .

- (b) (5 points) Find an equation of the plane through the point $R(5, 2, -2)$, and perpendicular to the planes $x + 3y - 2z = 11$ and $2x - y = 23$.

Solution: A normal vector to the plane $x + 3y - 2z = 11$ is $\langle 1, 3, -2 \rangle$. A normal vector to the plane $2x - y = 23$ is the vector $\langle 2, -1, 0 \rangle$. The plane we are looking for is perpendicular to these two planes. Hence any normal vector of this plane must be perpendicular to $\langle 1, 3, -2 \rangle$ and $\langle 2, -1, 0 \rangle$. To find such a vector we take the cross product

$$\mathbf{n} = \langle 1, 3, -2 \rangle \times \langle 2, -1, 0 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & -2 \\ 2 & -1 & 0 \end{vmatrix} = -2\mathbf{i} - 4\mathbf{j} - 7\mathbf{k} = \langle -2, -4, -7 \rangle.$$

Hence, an equation of the plane through $R(5, 2, -2)$ and with normal vector $\mathbf{n} = \langle -2, -4, -7 \rangle$ is

$$-2(x - 5) - 4(y - 2) - 7(z + 2) = 0,$$

which we can rewrite as $2x + 4y + 7z = 4$.

- (c) (5 points) Find the intersection between the line $x = 2 + t$, $y = -1 + 4t$, $z = 3 - 2t$ and the plane $x + 2y + 3z = 3$.

Solution: To find the intersection of the two we have to substitute the equation of the line in the equation of the plane:

$$\begin{aligned}(2 + t) + 2(-1 + 4t) + 3(3 - 2t) &= 3 \iff 2 + t - 2 + 8t + 9 - 6t = 3 \\ &\iff 3t = -6 \\ &\iff t = -2.\end{aligned}$$

For $t = -2$ we find the corresponding point on the line by substituting in the parametric equation

$$\begin{cases} x = 2 + 2 = 0 \\ y = -1 + 4 \cdot (-2) = -9 \\ z = 3 - 2 \cdot (-2) = 7. \end{cases}$$

In conclusion, the point of intersection between the line and the plane is $(0, -9, 7)$.

4. Consider the system

$$\begin{aligned}2x_2 + 7x_4 &= 0 \\ x_2 - x_3 + 2x_4 &= 0 \\ x_1 + 2x_3 &= 0\end{aligned}$$

- (a) (3 points) Without solving the system, explain why it cannot have a unique solution.

Solution: The system has $n = 4$ unknowns and $m = 3$ equations, hence the rank is $r \leq 3$. In particular, we can never have $r = n = 4$, which is required to have a unique solution.

- (b) (10 points) Use Gauss-Jordan elimination to find the solutions of the system. Indicate the rank of the system, the number of parameters of the system and list the basic solution(s) if they exist.

Solution: The augmented matrix of the system is

$$\left(\begin{array}{cccc|c} 0 & 2 & 0 & 7 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 1 & 0 & 2 & 0 & 0 \end{array} \right).$$

We bring the matrix in reduced row echelon by the following sequence of row operations

$$\begin{aligned}
 & \left(\begin{array}{cccc|c} 0 & 2 & 0 & 7 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 1 & 0 & 2 & 0 & 0 \end{array} \right) \begin{array}{l} R_1 \longrightarrow R_3 \\ R_3 \longrightarrow R_1 \end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 2 & 0 & 7 & 0 \end{array} \right) R_3 \longrightarrow R_3 - 2R_2 \\
 & \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 2 & 3 & 0 \end{array} \right) R_3 \longrightarrow R_3/2 \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 1 & 3/2 & 0 \end{array} \right) \begin{array}{l} R_1 \longrightarrow R_1 - 2R_3 \\ R_2 \longrightarrow R_2 + R_3 \end{array} \\
 & \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & -3 & 0 \\ 0 & 1 & 0 & 7/2 & 0 \\ 0 & 0 & 1 & 3/2 & 0 \end{array} \right)_{RREF}
 \end{aligned}$$

Translating this in equations gives $x_1 - 3x_4 = 0$, $x_2 + 7x_4/2 = 0$ and $x_3 + 3x_4/2 = 0$, equivalently

$$\begin{aligned}
 x_1 &= 3x_4 \\
 x_2 &= -7x_4/2 \\
 x_3 &= -3x_4/2.
 \end{aligned}$$

Setting $t = x_4$, we find that the general solution of the system is

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3t \\ -7t/2 \\ -3t/2 \\ t \end{pmatrix} = t \begin{pmatrix} 3 \\ -7/2 \\ -3/2 \\ 1 \end{pmatrix}.$$

Finally, the system has rank $r = 3$ and $n - r = 4 - 3 = 1$ parameter (denoted by t here). A basic solution is

$$\begin{pmatrix} 3 \\ -7/2 \\ -3/2 \\ 1 \end{pmatrix}.$$